

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 4

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TITLE

Develop Dynamic Webpage Using JavaScript

Aim

To develop a dynamic webpage using JavaScript that accepts student details and subject marks, calculates the total marks, average, grade, and result, and dynamically displays the results on the webpage.

OBJECTIVE

The objectives of this experiment are:

- To understand the basics of JavaScript and DOM manipulation.
 - To accept student details and subject marks through HTML input fields.
 - To retrieve input values using JavaScript.
 - To calculate total and average marks.
 - To determine the grade based on the average marks.
 - To determine whether the student has passed or failed.
 - To dynamically update the webpage using JavaScript.
 - To implement a Reset button for clearing the entered data.
 - To apply dynamic styling based on the grade.
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Theory

JavaScript can be used to make a webpage dynamic and interactive by responding to user actions and modifying HTML elements without reloading the page.

In this experiment, a Student Marks Calculator is developed using HTML, CSS, and JavaScript. The webpage accepts the student's name and marks in five subjects: English, Maths, Science, Social, and

SL.

When the Calculate button is clicked, JavaScript retrieves the values entered by the user and calculates the total and average marks. Based on the average, a grade is assigned from A to D or Fail. The program also checks whether the student has scored at least 35 marks in every subject to determine whether the student has passed or failed.

The webpage uses DOM manipulation and event listeners to dynamically update the result section. The grade color is also changed dynamically according to the grade. The Reset button clears all the entered values and resets the displayed results.

Main Features

- Accepts student's name.
- Accepts marks for five subjects.
- Calculates total marks.
- Calculates average marks.
- Assigns a grade based on average.
- Determines PASS or FAIL.
- Changes the grade color dynamically.
- Displays results without refreshing the webpage.
- Provides a Reset button to clear the form.

Concepts Used

JavaScript Concept	Purpose
getElementById()	Accesses HTML elements
addEventListener()	Handles button click events
Number()	Converts input values into numbers
if-else	Determines grade and result
textContent	Dynamically changes webpage text
.value	Gets or modifies input values
DOM Manipulation	Updates webpage elements dynamically
style.color	Changes the grade color dynamically

ALGORITHM / PROCEDURE

1. Create an HTML webpage containing input fields for the student's name and five subject marks.
2. Create Calculate and Reset buttons.

3. Create a result section to display the student name, total, average, grade, and result.
4. Apply CSS styling to the webpage and its components.
5. Create a JavaScript file and select the required HTML elements using getElementById().
6. Add a click event listener to the Calculate button.
7. Retrieve the student's name and subject marks using .value.
8. Convert the marks into numbers using Number().
9. Calculate the total marks.
10. Calculate the average by dividing the total by 5.
11. Determine the grade using if-else conditions.
12. Change the grade color according to the grade.
13. Check whether all subjects have at least 35 marks.
14. Display PASS if all subjects satisfy the minimum marks requirement; otherwise display FAIL.
15. Add a click event listener to the Reset button.
16. Clear all input fields and reset the displayed results.
17. Execute the application and verify the output using multiple test cases.

Code

HTML Code

```
</DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student grade calculator</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div>
    <div class="heading">
      <h1>Student Grade calculator</h1>
    </div>

    <div class="marks">
      <label for="" class="name">Name</label>
      <input type="text" placeholder="Enter the your name" class="ans" id="studentName">

      <br><br>
      <label for="" class="name">English</label>
      <input type="number" placeholder="Enter the marks" class="ans" id="eng">

      <br><br>
      <label for="" class="name">Maths</label>
      <input type="number" placeholder="Enter the marks" class="ans" id="math">

      <br><br>
      <label for="" class="name">Science</label>
      <input type="number" placeholder="Enter the marks" class="ans" id="sci">

      <br><br>
      <label for="" class="name">Social</label>
      <input type="number" placeholder="Enter the marks" class="ans" id="soc">

      <br><br>
      <label for="" class="name">SL</label>
      <input type="number" placeholder="Enter the marks" class="ans" id="sl">

      <br><br>
      <button class="btn" id="calculateBtn">Calculate</button>
      <button type="reset" class="btn" id="resetBtn">Reset</button>
    </div>

    <div class="results">
      <h3>Results: </h3>
      <p id="student">Name :</p>
      <p id="total">Total :</p>
      <p id="average">Average:</p>
      <p id="grade">Grade:</p>
      <p id="result">Result:</p>
    </div>
  </div>

  <script src="script.js"></script>
</body>
</html>
```

CSS Code:

```
body{
  /* display: flex; */
  background-color: blanchedalmond;
}
/* .heading{
  display: flex;
  justify-content: center;
} */
h1{
  /* display: flex; */
  /* justify-content: center; */
  display: flex;
  justify-content: center;
  margin: 0 auto 20px auto;
  /* border: 2px solid black; */
  background-color: white;
  width: 500px;
  flex-wrap: wrap;
  border-radius: 20px;
  padding: 10px;
  box-shadow: 0px 10px 20px rgba(0, 0, 0, 0.2);
}

.marks{
  background-color: rgb(199, 167, 230);
  margin: 80px;
  padding: 40px;
  border-radius: 15px;
  /* gap: 40px; */
  display: flex;
  flex-direction: column;
  justify-content: center;
  /* text-align: center; */
  gap: 10px;
}

.name{
  /* border: 2px solid black; */
  /* margin: 0 auto 10px auto; */
  color: rgb(19, 17, 173);
  font-size: 1rem;
}

.ans{
  border-radius: 5px;
  border-color: none;
  margin: 10px;
  padding: 8px;
  width: 250px;
}

.btn{
  width: 100px;
  display: flex;
  border-color: none;
  background-color: rgb(186, 186, 226);
  border-radius: 8px;
  color: white;
  padding: 8px;
  box-shadow: 0px 10px 20px rgba(0, 0, 0, 0.15);
}

.btn:hover{
  transform: scale(1.2);
  transition: 1s;
}
```

```
.results h3{
  color: white;
  display: flex;
  justify-content: center;
  font-size: 1.5rem;
}

.results p{
  color: rgb(72, 64, 190);
  font-weight: bold;
}
```

Javascript code:

```
const calculateBtn = document.getElementById("calculateBtn");
calculateBtn.addEventListener("click", function(){

const studentName = document.getElementById("studentName").value;
document.getElementById("student").textContent = `Name : ${studentName}`

const english = Number(document.getElementById("eng").value);
const maths = Number(document.getElementById("math").value);
const science = Number(document.getElementById("sci").value);
const social = Number(document.getElementById("soc").value);
const sl = Number(document.getElementById("sl").value);

const total = english + maths + science + social + sl;
console.log(total);

const totalElement = document.getElementById("total");
totalElement.textContent = `Total : ${total}`;

const average = total / 5;
const avgElement = document.getElementById("average");
avgElement.textContent = `Average: ${average}`;

let grade;
if(average >= 90){
    grade = "A";
}
else if(average >= 80){
    grade = "B";
}
else if(average >= 70){
    grade = "C";
}
else if(average >= 60){
    grade = "D";
}
else{
    grade = "Fail";
}

document.getElementById("grade").textContent = `grade : ${grade}`;

const gradeElement = document.getElementById("grade");
if(grade == "A"){
    gradeElement.style.color = "green";
}
else if(grade == "B"){
    gradeElement.style.color = "blue";
}
else if(grade == "C"){
    gradeElement.style.color = "yellow";
}
else if(grade == "D"){
    gradeElement.style.color = "orange";
}
else{
    gradeElement.style.color = "red";
}

let result;

if (
    english >= 35 &&
    maths >= 35 &&
```

```
let result;

if (
    english >= 35 &&
    maths >= 35 &&
    science >= 35 &&
    social >= 35 &&
    sl >= 35
) {
    result = "PASS";
}
else {
    result = "FAIL";
}

document.getElementById("result").textContent = `Result : ${result}`;
});

const resetBtn = document.getElementById("resetBtn");
resetBtn.addEventListener("click", function () {
    document.getElementById("studentName").value = "";
    document.getElementById("eng").value = "";
    document.getElementById("math").value = "";
    document.getElementById("sci").value = "";
    document.getElementById("soc").value = "";
    document.getElementById("sl").value = "";

    document.getElementById("student").textContent = "Name :";
    document.getElementById("total").textContent = "Total :";
    document.getElementById("average").textContent = "Average :";
    document.getElementById("grade").textContent = "Grade :";
    document.getElementById("result").textContent = "Result :";

    document.getElementById("grade").style.color = "";
});
```

CODE EXPLANATION

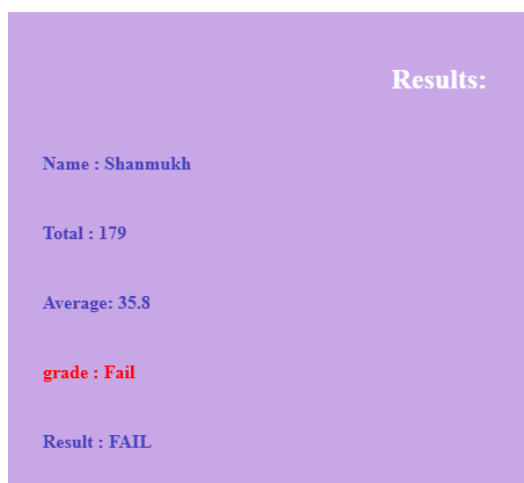
Code	Explanation
getElementById()	Selects an HTML element using its ID.
addEventListener()	Handles events such as button clicks.
.value	Retrieves the value entered in an input field.
Number()	Converts the entered marks from string values into numbers.
english + maths + science + social + sl	Calculates the total marks obtained in all five subjects.
total / 5	Calculates the average marks.
if-else	Determines the grade and pass/fail result based on conditions.

textContent	Dynamically updates the result displayed on the webpage.
style.color	Changes the grade color dynamically according to the grade.
>= 35	Checks whether the student has scored the minimum passing marks in each subject.
gradeElement	Stores the HTML element used to display the grade.
resetBtn	Selects the Reset button from the webpage.
.value = ""	Clears the values entered in the input fields.
innerText	Updates the displayed text such as the total number of contacts or results.
style.color = ""	Resets the dynamically applied grade color.
resetBtn.addEventListener()	Handles the click event of the Reset button.

TEST CASES AND EXECUTION DATA

Test Case	Component Tested	Action / Input	Expected Result	Actual Result	Status
TC-01	Calculate Total and Average	Enter Name: Vani and marks: 80, 90, 85, 75, 70 and click Calculate	Total and average should be calculated correctly	Total and average displayed correctly	Pass
TC-02	Grade Calculation	Enter marks with an average of 90 or above	Grade should be A and displayed in green	Grade A displayed in green	Pass
TC-03	Pass/Fail Result	Enter marks of at least 35 in all five subjects	Result should be PASS	PASS displayed successfully	Pass
TC-04	Fail Result	Enter less than 35 marks in any one subject	Result should be FAIL	FAIL displayed successfully	Pass
TC-05	Reset Function	Enter student details, calculate the result, and click Reset	All input fields and displayed results should be cleared	Fields and results cleared successfully	Pass

TC OUTPUT 04 :



Output :

Student Grade calculator

Name

English

Maths

Science

History

Art

Calculate

Reset

Results:

Total: 508

Average: 84.7

grade: **B**

Result: **PASS**

RESULT

Thus, a dynamic Student Marks Calculator was successfully developed using HTML, CSS, and JavaScript. The application successfully accepts student details and subject marks, calculates the total and average, determines the grade and PASS/FAIL result, dynamically changes the grade color, and resets the form using JavaScript DOM manipulation and event handling.

